

0.1 Why the constructed neural controls have invariant laws

The following argument concerns the explicitly forced, bounded-parameter models in Appendix ??; it does not assert recurrence for ordinary-network SGD.

Proposition 1 (Bounded data gradients and latent return). *On a finite-dimensional active parameter space, consider the exact update*

$$\theta^+ = a\theta + b_B(\theta) + \nu Z, \quad a = 1 - \eta\mu, \quad |a| < 1, \quad \sup_{B, \theta} \|b_B(\theta)\| \leq K < \infty, \quad Z \sim N(0, I), \quad \nu > 0.$$

Assume fresh iid batches and Gaussian vectors, independent of each other and of the current state, and continuous b_B . The chain has a unique attracting invariant probability, a smooth strictly positive density, and every finite polynomial moment. These conclusions apply to the constructed tanh/sigmoid networks, including their frozen-output-bias ablation.

Proof. First verify the bounded-gradient hypothesis for the stated networks. Let $d_0, \dots, d_L$, with $d_L = 1$, be fixed layer widths, and write $W_\ell = w_\ell \tanh \Theta_\ell$, $b_\ell = \tanh v_\ell$, $w_\ell = 2/\sqrt{d_{\ell-1}}$. Finite inputs and targets give bounds X_*, Y_* . Hidden activations and their derivatives have absolute value at most one. Thus hidden outputs satisfy $A_\ell \leq 1$, with $A_0 = X_*$, and the linear output satisfies $|f_\theta(x_i)| \leq F = d_{L-1}w_L + 1$. Set $D_L = 1$ and $D_\ell = d_{\ell+1}w_{\ell+1}D_{\ell+1}$. Backpropagation and $|\tanh'| \leq 1$ give, for every latent coordinate,

$$|\partial_{\Theta_{\ell,jk}} f_\theta| \leq D_\ell w_\ell A_{\ell-1}, \quad |\partial_{v_{\ell,j}} f_\theta| \leq D_\ell.$$

There are finitely many coordinates, so $\|\nabla f_\theta\| \leq J < \infty$ uniformly. Hence every individual and averaged squared-loss data gradient has norm at most $G = (F + Y_*)J$. The sampled objective therefore gives $b_B = -\eta g_B$, $K = \eta G$, and $\nu = \eta \sigma_{\text{eff}} > 0$ exactly.

Put $q = |a|$ and $V = 1 + \|\theta\|$. The triangle inequality gives $PV \leq qV + 1 - q + K + \nu \mathbb{E}\|Z\|$. On $\|\theta\| \leq R$, all conditional Gaussian means have norm at most $qR + K$. Consequently their mixture density, on a target ball $\|u\| \leq r$, is bounded below by

$$(2\pi\nu^2)^{-P/2} \exp[-(r + qR + K)^2/(2\nu^2)] > 0,$$

where P is the active dimension. This gives compact minorization, Lebesgue irreducibility, and aperiodicity. Localizing the displayed geometric drift to a sufficiently large ball, the Foster–Harris theorem gives uniqueness and weighted total-variation convergence (?). For each $p > 0$, subadditivity ($p \leq 1$) or Young’s inequality ($p > 1$) gives $P\|\theta\|^p \leq q_p\|\theta\|^p + C_p$, with $q_p < 1$. Iteration from a finite start, convergence in law, and Portmanteau give the invariant p th moment. Invariance expresses its density as a mixture of translated Gaussians with covariance $\nu^2 I$; uniformly bounded Gaussian derivatives justify differentiation and imply smoothness and positivity. Freezing a bias removes its coordinate: the same argument holds on the remaining space, where the forcing is still full rank. $\square$

For the reported $\mu = .05$ and $\eta \in \{.5, 10\}$, the outer factors are .975 and .5. The proof establishes recurrence independently of any measured tangent-growth sign; it proves neither positive Lyapunov growth nor a density mode count.